Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/24/2017
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Project Title:	Coherent Magnon Optics		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL)
Start Date:	04/01/2014	**End Date:**	04/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	6	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mueller, Holger Ph.D. (University of California, Berkeley)		
Grant/Contract No.:	JPL 1505588		
Performance Goal No.:			
Performance Goal Text:			

The spin degree of freedom of ultracold atoms makes the physics of quantum gases richer, and offers opportunities for fundamental science explorations using cold atomic media. The scientific agenda we propose for the Cold Atom Laboratory (CAL) builds upon on our world-leading research on spinor Bose-Einstein condensates, including our development of magnetization-sensitive imaging and studies of spin coherence in optically trapped quantum gases. We propose four activities:

(1) Precise characterization of magnons in spinor Bose-Einstein condensates: Magnons are low energy spin excitations in spinor condensates. Theories of symmetry breaking, as well as mean-field and beyond-mean-field mean-field treatments of Bose gases provide strong predictions that we will test precisely with interferometric measurements based on coherent magnon optics.

Task Description:	(2) Magnon and Bragg interferometry in dense and dilute Bose condensed gases: magnons in ferromagnetic spinor condensates, such as the $F=1$ gases of 87Rb and 41K, are by necessity gapless and quadratically dispersing, with an effective mass that, within mean field theory that is applicable to weakly interacting gases, is equal to the atomic mass. In other words, magnons propagate just like free particles. We propose to implement an atom interferometer based on coherent magnon optics. Compared to a Bragg interferometer, a magnon interferometer has no mean-field interaction shift. Comparing the two interferometers provides a strong measurement cross-check and may allow quantitative studies of polaronic (quantum depletion) energy shifts. (3) Imaging superflow with a magnon shear interferometer: We propose to image the magnon interference pattern to perform in-situ measurements of superfluid flow in three dimensions, akin to particle tracking velocimetry. We propose to utilize the CAL to expand a trapped Bose-Einstein condensate into a large three dimensional volume, essentially magnifying the superfluid flow pattern sufficiently that it can be imaged. A three-dimensional spatial pattern of magnons will be allowed to propagate coherently before being imaged, revealing the three-dimensional superfluid velocity field. (4) Magnon-evaporative cooling: We will explore a novel cooling technique in which magnon excitations are used to absorb, and then expel entropy from a highly degenerate spinor Bose-Einstein condensate. Our target is to produce quantum gases at the lowest entropy per particle, with a method that can be applied in the CAL apparatus.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is fundamental in flavor. We are exploring new phenomena and new experimental capabilities in a physical system that is itself new and only recently available for experimentation. Exploring in this new space of questions has prospects for making discoveries. More specifically, the research impacts our knowledge of materials (through studies of superfluid turbulence), extends the frontiers of natural phenomena that we can access and explore (through the development of new thermometry and cooling techniques), and contributes to the development of high precision sensors for future scientific and technological applications (through studies of atom and magnon interferometry).
Task Progress:	This narrative describes progress on this project during the period 4/1/2016 – 12/31/2016. We report on three activities: 1. Quantum gas mixtures of Li and Rb: In the previous reporting period, we began augmenting our experimental capabilities by adding ultracold lithium gases into our experiment. The goal of this activity is to study spinor Bose-Einstein gases and also magnon interferometry with a spinor Bose gas that has different properties than does rubidium. Compared with rubidium, lithium has the property that it should equilibrate more rapidly owing to a stronger spin-dependent interaction. We also are interested in preparing quantum-gas mixtures in optical lattices, and gases of ultracold dipolar molecules for applications in precision measurement, quantum dynamics, ultracold chemistry and quantum simulation. Our experimental efforts during this reporting period were focused on this task. We built a laser system that delivers light at several different frequencies as are needed for slowing and laser cooling the lithium-7 isotope. We adapted our vacuum system, modifying the effusive oven source so that it emits an atomic beam containing both rubidium and lithium atoms. Operating this two-element oven adds a significant gas load to the chamber, both from the atomic beam itself and also from the additional outgassing of vacuum components at the high temperatures needed for the oven to operate. We diagnosed this higher gas load and modified the vacuum chamber to allow for ultrahigh vacuum conditions still. We developed a method for loading both rubidium and lithium atoms into the same magneto-optical trap. We found that operating the two traps simultaneous caused light-induced losses wherein lithium and rubidium atoms collide in the presence of light near the rubidium resonance. In such a collision, a red-detuned photon is emitted, leaving behind energy that is taken up as kinetic energy by the atoms, causing both to be lost from the trap. After measuring the loss rate from such collisions, finding them to be consistent with previous measurements, we adopted a loading sequence in which the lithium and rubidium atoms accumulate in the trap while imbalanced light forces are used to offset one element from the other spatially. This arrangement allows the populations of both elements to build up. Near the end of the loading sequence, we restore the balance of optical forces on the atoms, causing the trapped gases to overlap, but for a sufficiently short time that the losses are not severe. We implemented a recently suggested form of laser cooling to the lithium gas. In most prior experiments, laser cooling of lithium ends up being limited at the Doppler temperature because the excited-state structure of lithium prevents the atoms from being cooled by Sisyphus cooling. Thus, magneto-optical traps of lithium deliver atoms at temperatures of hundreds of microkelvin, much hotter than is achieved with other laser cooled elements. However, it was recently shown that a form of dark-state laser cooling can be applied, making use of the resolved structure of the D1 resonance line of lithium. We implemented this “D1 laser cooling” to cool all three dimensions of motion of lithium, achieving temperatures in the 10s of microkelvin range. We have also succeeded in loading the two elements into a magnetic trap. The magnetic trap lifetime for both species is very long – on the order of two minutes – limited by the residual gas in the ultrahigh vacuum chamber. At present, we are studying the process of thermal equilibration. 2. Considering experimental tests of the validity of quantum mechanics on macroscopic scales: One of the major motivations for conducting experiments on ultracold atoms in free fall, is to perform atom interferometry experiments in which matter waves are separated by large distances for long periods of time. Increasing both quantities improves the sensitivity of atom interferometer to acceleration, perhaps permitting highly sensitive tests of the gravitational equivalence principle or measurements of gravitational waves. The microgravity environment of space-borne experiments allows interferometry experiments with freely falling atoms at larger spatial and temporal separation. In contrast, on Earth, the free-fall time of the atoms is limited by gravity. Stretching the bounds of atom interferometry also tests our assumption that quantum mechanics continues to be valid at ever-larger length and timescales. Several theories have been offered that consider the possibility that quantum mechanics does not retain its validity under such macroscopic separations, citing the empirical fact that macroscopic

objects appear to be described perfectly well by classical, rather than quantum, mechanics. One such theory is the theory of continuous spontaneous localization (CSL), which posits that massive objects that are placed in superpositions of spatially separated states will decohere naturally. While it has been experimentally verified that massless particles (photons) can remain in coherent superpositions with macroscopic length and time scales, it remains untested whether massive particles can retain coherence under such conditions.

A recent paper by the Kasevich group at Stanford has claimed to test the validity of quantum mechanics for rubidium atoms placed in superpositions with meter-scale spatial and second-scale temporal separations. However, as described in our publication [Dan M. Stamper-Kurn, G. Edward Marti, Holger Müller, “Brief Communication Arising: Verifying quantum superpositions at metre scales,” Nature 537, http://dx.doi.org/ (2016)], the Stanford group’s interpretation is flawed. In their experiment, they observe high contrast interference for two Bose-Einstein condensed gases that are separated by large distances and large times. However, the interference phase is not stable between measurements. The instability of the interference phase is, in fact, completely consistent with spontaneous localization of the spatially separated clouds, causing them to interfere just like two independently produced, and entire phase-incoherent, Bose-Einstein condensates. We show that their experiment sets only a modest bound on spontaneous localization theories – at spatial separations of just a millimeter scale – matching bounds that had already been set by prior experiments. Our correct interpretation of their work is highly significant for future efforts to test CSL theories with space-borne experiments, i.e., we show that the validity quantum mechanics at macroscopic scales remains untested.

3. Defining future space-based ultracold atom experiments: NASA has begun discussions with the German space agency (DLR) about a joint mission that will be a follow-on to the Cold Atom Laboratory. I contributed to these discussions by attending a joint planning workshop in Bremen, Germany in December 2016. I presented a scientific proposal to use a new mission to study ultracold atomic gases in large volume, blue-detuned optical traps. Such a capability would allow one to study the dynamics and thermodynamics of homogeneous quantum gases. I contributed to the planning documentation. The ideas I brought to the meeting do not constitute a recognized scientific direction for the proposed mission.

Bibliography Type:	Description: (Last Updated: 06/16/2021)
Articles in Peer-reviewed Journals	Stamper-Kurn DM, Marti GE, Müller H. "Brief Communication Arising: Verifying quantum superpositions at metre scales." 2016 Sep 1;537(7618):E1-2. http://dx.doi.org/10.1038/nature19108 ; PubMed PMID: 27582225 , Sep-2016